Stationary states of the selected Slater algebra

A compact-ideal classification and the covariant-observable interface

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Abstract

The same-disorder Slater transfer has a bounded, purely absolutely continuous Hamiltonian and a compact selected creation operator. We identify its norm algebra as continuous functions of the Hamiltonian plus compact operators. Every stationary state on this algebra factors through its commutative energy quotient and annihilates all selected insertions. There is no normal stationary density on the full Type I closure. We construct finite-time averaged normal states and quantify their loss of finite-rank signal. A separate exact covariance argument recovers tracial correlators on microscopic covariant observables, providing the next route to nontrivial stationary bilocal dynamics. The conclusions concern a specified algebra, representation and flow; they are not a no-go statement for de Sitter holography.

1 The physical question and existing input

Rahman–Susskind distinguish Boltzmann temperature from Hawking temperature and tomperature, with explicit cosmic/string unit conversions [3]. We therefore do not infer a Gibbs density in our representation from a de Sitter temperature. Rajgadia–Xu distinguish generic probes from state-adapted dressed probes and obtain a Type I algebra for the latter [2, Secs. 4.2, 9.2]. The present same-J complex model differs from their matter ensemble. Our task is to examine the state space of the transfer already derived.

Normal states, invariance and the ordinary Type I Gibbs criterion are standard [1, Secs. 8.3–8.6]. The compactness arguments below are elementary applications, written out to separate normal states, singular states and changes of observable algebra.

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2 Model, units and the norm algebra

Use the established half-filled Slater construction with standard CAR, overlapping p-body sets and the same Hermitian Gaussian J:

$$H_N = \sqrt{w_N} \sum_{|A|=|B|=p} J_{AB} C_A^\dagger C_B, \qquad \mathbb{E} J_{AB} J_{CD} = \delta_{AD}\delta_{BC}, \tag{1}$$

$$M_N^\dagger = L_N^{-1/2} \sum_{A\subset O^c, B\subset O} J_{AB} C_A^\dagger C_B, \qquad h_N = H_N/\sqrt{w_N D_N}, \tag{2}$$

$$L_N = \binom{N/2}{p}^2, \qquad D_N = \binom{N/2}{p}\binom{N/2+p}{p}. \tag{3}$$

The state is $\varphi_N = \mathbb{E}\langle O| \cdot |O\rangle$. Take $p^2/N \to \lambda > 0$, $q = e^{-4\lambda} \in (0,1)$ and $\hbar = 1$. Energies below are dimensionless h energies; physical string energies multiply by $J_s/\sqrt{\lambda}$. Times and inverse temperatures are conjugate to h, not cosmic quantities. All long-time statements follow the double-scaled limit; no uniform finite-N late-time limit is assumed.

The preceding transfer stage supplies a separable graded GNS space, a cyclic unit vector Ω, finite-dimensional levels, and

$$h = J_q \otimes I_{\mathcal{P}}, \qquad J_q|n\rangle = \sqrt{[n+1]_q}|n+1\rangle + \sqrt{[n]_q}|n-1\rangle, \tag{4}$$

$$M \text{ compact}, \qquad Q = MM^\dagger - qM^\dagger M = q^{\mathcal{N}}, \qquad Q^r \longrightarrow P_\Omega \quad \text{in norm.} \tag{5}$$

Here $\mathcal{P}$ is the primitive multiplicity space. The standard q-Hermite measure μ_q is absolutely continuous, positive in the interior of $I = [-E_*, E_*]$, $E_* = 2/\sqrt{1-q}$. Thus h is purely absolutely continuous on the full joint space, irrespective of primitive multiplicity. Set $\mathcal{A} = C^*(1, h, M)$.

Proposition 1. *$\mathcal{A} = C^*(h) + \mathcal{K}(\mathcal{H})$, with unique decompositions $f(h) + K$, and $\mathcal{A}/\mathcal{K}(\mathcal{H}) \simeq C(I)$ by $f(h) + K \mapsto f$.*

Proof. $P_\Omega \in \mathcal{A}$ by the norm limit above. Products $AP_\Omega B^*$ with $A, B \in \mathcal{A}$ and cyclicity norm-approximate every rank-one operator, so $\mathcal{K}(\mathcal{H}) \subset \mathcal{A}$. Conversely $M \in \mathcal{K}(\mathcal{H})$. Every nonempty relatively open spectral interval has an infinite-dimensional spectral projection. If $f \neq 0$ is continuous, $f(h)$ is bounded below on one such subspace and cannot be compact. More precisely its essential norm is $\|f\|_\infty$. The image of $C^*(h)$ in the Calkin algebra is consequently isometric and closed. Its inverse image is the closed sum $C^*(h) + \mathcal{K}(\mathcal{H})$, proving the assertion. □

3 Normal states and stationary quotient states

Let $\alpha_t(A) = e^{ith}Ae^{-ith}$. It is norm-continuous since h is bounded, and preserves $\mathcal{A}$ and its compact ideal.

Proposition 2. *There is no normal stationary state on $B(\mathcal{H})$ for α. For every finite real β, $e^{-\beta h}$ is not trace class.*

Proof. A normal stationary state has a positive trace-one density ρ commuting with h [1]. A nonzero positive trace-class operator has a positive eigenvalue with finite-dimensional eigenspace. That eigenspace is h-invariant. The restriction of self-adjoint h to it has an eigenvector, contradicting the absence of point spectrum. Also $e^{-\beta h} \geq e^{-|\beta|E_*}I$ on an infinite-dimensional space, so its trace diverges. □

This does not exclude singular stationary states. Their restrictions to the norm algebra can instead be classified exactly.

Proposition 3. *Every stationary state of $\mathcal{A}$ is uniquely*

$$\omega_\nu(f(h) + K) = \int_I f(E)\mathrm{d}\nu(E), \qquad \nu \text{ a probability measure on } I. \tag{6}$$

Conversely every such state is stationary. All are tracial and are β-KMS for every finite $\beta > 0$ (and traces at $\beta = 0$).

Proof. The restriction of a positive bounded functional to $\mathcal{K}(\mathcal{H})$ has the form $K \mapsto \mathrm{Tr}(\rho_c K)$ with positive trace-class ρ_c, possibly of trace less than one. Stationarity implies $e^{-ith}\rho_c e^{ith} = \rho_c$, by testing all rank-one K. The previous eigenspace argument forces $\rho_c = 0$. The state therefore factors through the quotient in Proposition 1; the Riesz representation theorem gives (6). The quotient dynamics is the identity and the quotient is commutative, so these states are tracial and obey the KMS boundary condition, with a constant analytic function. □

In particular $\omega_\nu(M^\dagger M) = \omega_\nu(MM^\dagger) = 0$. Every finite product of bounded elements of $\mathcal{A}$ that contains an M or $M^\dagger$ insertion is compact, including time-translated insertions. Its stationary expectation is zero. The GNS algebra of ω_ν is the abelian multiplication algebra $L^\infty(I, \nu)$. The energy distribution is arbitrary; KMS alone does not choose $e^{-\beta E}\mu_q/Z$ here. This loss of dynamical information is stronger than merely failing to normalize an ordinary trace.

4 Weak closure and long-time preparation

The von Neumann closure of $\mathcal{A}$ is $B(\mathcal{H})$, but stationary states of the norm algebra must not be confused with normal states of that closure. There is also no singular finite-β KMS state on the full $B(\mathcal{H})$ for this bounded inner flow. Indeed, if ω were one, h lies in its centralizer and

$$\tau(A) = \frac{\omega(e^{\beta h}A)}{\omega(e^{\beta h})} \tag{7}$$

would be a positive normalized trace. Positivity follows by moving $e^{\beta h/2}$ through the functional. The KMS identity $\omega(XY) = \omega(Y\alpha_{i\beta}(X))$ gives $\tau(AB) = \tau(BA)$. Two isometries V_1, V_2 with orthogonal ranges summing to I would then give $1 = \sum_i \tau(V_i V_i^*) = \sum_i \tau(V_i^* V_i) = 2$. This rules out KMS states on the full closure, not arbitrary singular invariant states. It also explains why a KMS state on $\mathcal{A}$ cannot extend as a KMS state to that closure.

Proposition 4. *For every normal density ρ, as $|t| \to \infty$,*

$$\mathrm{Tr}(\rho\alpha_t(f(h)+K)) \longrightarrow \int_I f(E)\mathrm{d}\nu_\rho(E), \qquad \nu_\rho(B) = \mathrm{Tr}(\rho P_h(B)). \tag{8}$$

This is pointwise weak- convergence on $\mathcal{A}$.*

Proof. The scalar spectral measure between any two vectors has an L^1 density, since h is purely absolutely continuous. The Riemann–Lebesgue lemma gives $e^{-ith}\psi \rightharpoonup 0$. For compact K, $Ke^{-ith}\psi \to 0$ in norm; hence its vector expectation vanishes. Trace-norm finite-rank approximation of ρ extends this to all normal densities. The $f(h)$ part is conserved exactly. This proves (8) without time averaging. □

For the previous selected Euclidean preparation $\psi_\beta = Z_M^{-1/2} e^{-\beta h/2} M^\dagger \Omega$, this yields

$$\mathrm{d}\nu_{\psi_\beta}(E) = \frac{e^{-\beta E}(1 - \sqrt{q} + \sqrt{q}E^2)}{Z_M}\mathrm{d}\mu_q(E). \tag{9}$$

Its energy bias survives, while compact selected signals disappear. The limit has no density operator in this representation. Nor is convergence claimed for every bounded observable of $B(\mathcal{H})$. The inverse dressing $q^{-\mathcal{N}/2}M$ is unbounded and is outside the compact-ideal argument; it cannot be inserted into these bounded-word statements without new domain and moment control.

5 Finite observation times: an explicit family

Normal approximately stationary states do exist at finite duration. For $T > 0$ define the trace-norm integral

$$\rho_T = \frac{1}{2T} \int_{-T}^{T} e^{-ith} P_\Omega e^{ith} \mathrm{d}t. \tag{10}$$

It is positive of trace one. Comparing two shifted intervals gives

$$\left\| e^{-iuh} \rho_T e^{iuh} - \rho_T \right\|_1 \leq \min\{2, |u|/T\}. \tag{11}$$

The bound follows from the measure of the symmetric difference; it requires no spectral approximation. Write $R(t) = \langle \Omega, e^{ith}\Omega \rangle = \int e^{itE} \mathrm{d}\mu_q(E)$. Then, exactly in the limiting model,

$$\mathrm{Tr}(\rho_T P_\Omega) = \frac{1}{T} \int_0^T |R(t)|^2 \mathrm{d}t, \tag{12}$$

$$\mathrm{Tr}(\rho_T^2) = \frac{1}{T} \int_0^{2T} \left(1 - \frac{t}{2T}\right) |R(t)|^2 \mathrm{d}t. \tag{13}$$

The second identity follows by integrating over the difference of the two times in $[-T, T]^2$. The q-Hermite density $r_q(E)$ is bounded on a compact interval for each fixed $q < 1$, so $r_q \in L^2$. Parseval and dominated convergence give

$$\kappa_q = \pi \int_I r_q(E)^2 \mathrm{d}E, \qquad \mathrm{Tr}(\rho_T P_\Omega) \sim \frac{\kappa_q}{T}, \quad \mathrm{Tr}(\rho_T^2) \sim \frac{\kappa_q}{T}. \tag{14}$$

Thus smaller stationarity defect in (11) accompanies loss of any fixed finite-rank record, rather than convergence to a trace-class equilibrium. These are sequential limits, first double scaling and then $T \to \infty$.

For reproducible finite quadrature let E_j, w_j be the eigenvalues and vacuum weights of an L-site q-Jacobi matrix. Equations (12)–(13) become

$$S_L(T) = \sum_{jk} w_j w_k \,\mathrm{sinc}(T(E_j - E_k)), \tag{15}$$

$$P_L(T) = \sum_{jk} w_j w_k \,\mathrm{sinc}^2(T(E_j - E_k)), \qquad \mathrm{sinc}(x) = \sin x / x. \tag{16}$$

At fixed L their $T \to \infty$ limits are both $\sum_j w_j^2 > 0$. This is a discrete quadrature recurrence/pinching effect, not the nonzero purity of a continuum equilibrium. No finite-L plateau is used to prove (14).

6 A constructive return to covariant observables

The failure of selected stationary correlators does not remove the original bilocal route. Let $\Gamma_m(U) = \wedge^m U$ and let $B_a(J)$ be jointly $U(N)$-covariant, number-preserving random operators independent of the fixed reference occupation set O:

$$B_a(U \cdot J) = \Gamma_m(U) B_a(J) \Gamma_m(U)^*. \tag{17}$$

The Gaussian law is invariant under the induced action on its p-body indices. For every integrable word $F(J)$ in this family, possibly including h_N and fixed time evolutions, Schur's lemma on irreducible $\wedge^m \mathbb{C}^N$ gives

$$\mathbb{E}F(J) = \mathbb{E}\tau_m(F) I_m, \qquad \mathbb{E}\langle O|F(J)|O\rangle = \mathbb{E}\tau_m(F(J)), \quad \tau_m = \binom{N}{m}^{-1} \mathrm{Tr}_m \,. \tag{18}$$

This extends the H-only equality recorded in the earlier Slater stage; it is a direct symmetry argument, not a new general theorem. Products remain covariant, so the equality includes all joint moments rather than only two-point functions.

The restricted annealed state is tracial and stationary on this covariant algebra at finite N. For integrable Gibbs sandwiches, (18) also gives

$$\frac{\mathbb{E}\langle O|e^{-\beta h_N/2} F e^{-\beta h_N/2}|O\rangle}{\mathbb{E}\langle O|e^{-\beta h_N}|O\rangle} = \frac{\mathbb{E}\tau_m(e^{-\beta h_N} F)}{\mathbb{E}\tau_m(e^{-\beta h_N})}. \tag{19}$$

This is joint annealed normalization, not a disorder average of samplewise normalized Gibbs states. Finite-N cyclicity gives the KMS boundary relation. Passage to infinite families requires the existing moment/time-tail estimates for the actual observables. Equation (18) is a fixed-sector equality, not a license to replace the central sector by the full charge sum.

Fixed-O M_N is excluded from this covariance family. Its suppression in the old tracial state is already visible in

$$\mathbb{E}\,\mathrm{tr}_{\mathrm{full}}(M_N^\dagger M_N) = 4^{-p}, \quad \mathbb{E}\tau_{N/2}(M_N^\dagger M_N) = \frac{\binom{N-2p}{N/2-p}}{\binom{N}{N/2}} = 4^{-p}(1 + o(1)). \tag{20}$$

Each summand after Gaussian contraction is an occupancy projector on $2p$ distinct sites. The ratio follows from counting its allowed occupations. Expanding falling factorials gives the relative correction $O(p/N + p^3/N^2)$ in the stated double scaling. The full-trace identity was established previously; the saved small-system enumeration here checks both normalizations.

7 What a joint enlargement must accomplish

The covariant family provides a nontrivial stationary candidate without forcing the selected operator itself to be thermal. One should construct its joint GNS space with the same microscopic M_N, rather than append an unrelated independent matter probe. The old h, M cyclic space embeds as a reducing subspace for those generators whenever the joint limit exists. Compactness of M on that subspace does not imply compactness on the enlarged space; infinite new multiplicity can change it. Thus the present classification does not rule out this enlargement.

A useful diagnostic is sharper. Suppose $\mathcal{B} \subset \mathcal{A}$ contains $C^*(h)$ and is invariant under α, and a normal state ρ restricts to a stationary state on $\mathcal{B}$. For $B = f(h) + K \in \mathcal{B}$, also $K \in \mathcal{B}$. Stationarity and Proposition 4 applied to K^*K imply $\mathrm{Tr}(\rho K^*K) = 0$. Applied to KB' for $B' \in \mathcal{B}$, the same reasoning shows that the entire compact part vanishes in the restricted GNS representation. That representation is abelian with trivial flow. A covariant family with nonzero dynamical or noncommutative correlators cannot be represented this way inside the current norm algebra with the same normal state. This is a concrete test for the proposed bilocal interface.

CLPW instead change the algebra by adding an observer, imposing the constraint and selecting a positive-energy corner of a crossed product; their finite trace is not an ordinary trace on our $B(\mathcal{H})$ [4, Sec. 2.4]. We do not identify the two constructions. The present result motivates first completing the microscopic covariant joint interface within the Susskind program, before adding further observer structures.

The next calculation should retain one previously constructed nontrivial covariant bilocal generator B_N and evaluate its same-J mixed Gram data with h_N, M_N, beginning with

$$\mathbb{E}\langle O|M_N B_N^* B_N M_N^\dagger|O\rangle, \qquad \mathbb{E}\langle O|B_N^* M_N^\dagger M_N B_N|O\rangle. \tag{21}$$

Projection onto the existing h, M cyclic sector and the norm of the residual will determine whether new sectors are required. These quantities are research targets, not computed claims here. The microscopic interpretation of the joint grading, spatial and mass generators, finite-G interactions and horizon entropy remain open. The present stage fixes the state/algebra choice that those computations must respect.

8 Diagnostics and reproducibility

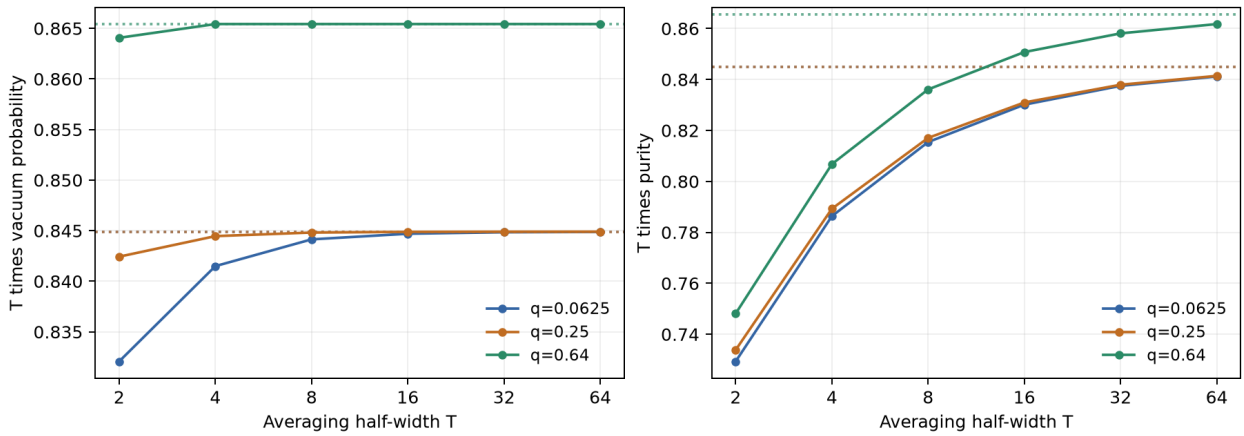


Figure 1: Finite-time normal states. Dotted levels are the independently integrated κ_q. Curves use 1100-node quadrature; the theorem behind the limiting behavior is Parseval, not the numerical fit.

The new driver uses $q = 0.0625, 0.25, 0.64$, Jacobi sizes $800, 1100$ and $T = 2, 4, 8, 16, 32, 64$, giving 36 quadrature rows. The largest size-comparison difference is 1.23×10^{-15}. Independent density integrations use 8193/16385 angle points and 256/384 product factors. They give approximately

$$(\kappa_{0.0625}, \kappa_{0.25}, \kappa_{0.64}) = (0.8448905756,\ 0.8449063017,\ 0.8654266760). \tag{22}$$

Three direct time-integral comparisons check the sinc and triangular-kernel formulas. Five finite occupation systems check the full/central trace factors by enumerating projectors. The output records relative asymptotic ratios, signed-sum conditioning, discrete plateaus, parameters, versions and input hashes. These are float64 diagnostics and exact finite counting, not interval certification or proofs of the infinite-dimensional propositions. No old scientific generator was rerun.

References

[1] J. Dereziński, *Introduction to Representations of the Canonical Commutation and Anticommutation Relations*, arXiv:math-ph/0511030v2, Secs. 8.3–8.6.

[2] H. Rajgadia and J. Xu, *Emergent States and Algebras from the Double-Scaling limit of Pure States in SYK*, arXiv:2604.14387v3.

[3] A. Rahman and L. Susskind, *Infinite Temperature is Not So Infinite: The Many Temperatures of de Sitter Space*, arXiv:2401.08555v3.

[4] V. Chandrasekaran, R. Longo, G. Penington and E. Witten, *An Algebra of Observables for de Sitter Space*, arXiv:2206.10780v3.